

11.1 Atoms, Nuclei & Radiation

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.1 Atoms, Nuclei & Radiation
Difficulty	Easy

Time allowed: 20

Score: /15

Percentage: /100

Question 1

A Geiger-muller tube held at a distance of about 12 cm from a radioactive source. The radiation emitted from the nuclear isotope was entirely stopped by a 2 mm thick sheet of lead.

Which of the statements can be deduced from the above information about the emission from the isotope?

- A. it could be alpha, beta and gamma radiation
- B. it could be beta and gamma radiation, but not alpha radiation
- C. it could be alpha and gamma radiation, but not beta radiation
- D. it could be alpha and beta radiation, but not gamma radiation

[1 mark]

Question 2

When β^- particles are emitted from identical nuclei in a substance, which of the following remain constant?

- A. proton number of the nuclei
- B. nucleon number of the nuclei
- C. neutron number of the nuclei
- D. energy of the β -particles

[1 mark]

Question 3

New insights were discovered when α -particles were fired at a sheet of gold foil.

Which conclusion can be drawn from the results of the experiment?

- A. the atomic nucleus contains protons and neutrons
- B. electrons orbit the atomic nucleus
- C. some atoms of the same element contain different numbers of neutrons
- D. atomic nuclei occupy a very small fraction of the volume of an atom

[1 mark]

Question 4

When aluminium ${}_{13}^{27}\text{Al}$ was bombarded with α -particles the first radioactive substance was made. The product of this reaction was an unstable isotope of phosphorus, ${}_{15}^{30}\text{P}$.

Which of the following is a by-product of this reaction?

- A. a neutron
- B. a γ -ray
- C. a β -particle
- D. an α -particle

[1 mark]**Question 5**

The atom ${}_{55}^{133}\text{Cs}$ is a neutral atom

How many protons, neutrons and electrons are in this atom?

	Protons	Neutrons	Electrons
A	133	55	133
B	55	78	55
C	78	55	78
D	55	133	55

[1 mark]**Question 6**

The following statements are about α -particles.

Which is correct?

- A. α -particles emitted from a single radioactive isotope have a continuous distribution of energies
- B. α -particles have less ionising power than β -particles
- C. the speeds of α -particles can be as high as $1.5 \times 10^7 \text{ m s}^{-1}$
- D. the charge of an α -particle is $+1.60 \times 10^{-19} \text{ C}$

[1 mark]

Question 7

Which of the following statements about the nucleus of the atom is correct?

- A. nucleon numbers are unchanged by the emission of β -particles
- B. in some nuclear processes, mass-energy is not conserved
- C. for some nuclei, the nucleon number can be less than the proton number
- D. different isotopic nuclei have different proton numbers

[1 mark]

Question 8

When an α -particle is produced by ${}_{92}^{258}\text{U}$ it produces a new atom X.

What are the values of the proton number and nucleon number for atom X?

	Protons	Nucleons
A	92	256
B	90	254
C	92	250
D	88	254

[1 mark]

Question 9

A nucleus Q decays into a nucleus R by emitting an alpha particle followed by two beta particles.

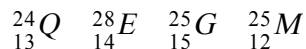
Which statement about this nuclear decay is correct?

- A. nucleus R is an isotope of nucleus Q
- B. nucleus R has the same nucleon number as nucleus Q
- C. beta particle decay occurs when a proton changes into a neutron
- D. the total mass of the products is equal to the mass of the initial nucleus Q

[1 mark]

Question 10

Four nuclei are represented below



Which of the following statements about these nuclei is correct?

- A. an uncharged atom of element Q has 24 orbital electrons
- B. when E absorbs a neutron and then emits an alpha particle, nucleus E transforms into M
- C. nuclei G and M are isotopes of the same element
- D. nucleus M could transform into Q by emitting a beta particle

[1 mark]

Question 11

Most α -particles pass straight through when fired at a thin sheet of metal foil, but a few are deflected by a large angle.

Which change would increase the **proportion** of α -particles deflected by a large angle?

- A. using a double thickness foil
- B. using a foil made of a metal with fewer protons in its nuclei
- C. using α -particles with greater kinetic energy
- D. using an alpha source with a higher activity

[1 mark]

Question 12

Plutonium-239 (${}_{94}^{239}\text{Pu}$) decays by emitting α -radiation.

Which nuclide is formed from one of these decay reactions? (The product nuclides are represented by X.)

- A. ${}_{95}^{239}\text{X}$
- B. ${}_{92}^{237}\text{X}$
- C. ${}_{93}^{239}\text{X}$
- D. ${}_{92}^{235}\text{X}$

[1 mark]

Question 13

One of the rows in the table shows a pair of nuclei that are isotopes of one another.

Which row is it?

	Nucleon number	Number of neutrons
A	186	112
	180	118
B	184	110
	187	110
C	186	112
	182	108
D	186	110
	186	112

[1 mark]

Question 14

Which statement about alpha, beta and gamma radiation is correct?

- A. gamma radiation has the greatest ionising power
- B. beta radiation has the greatest ionising power
- C. alpha radiation has the greatest ionising power
- D. alpha, beta and gamma radiation have nearly equal ionising powers

[1 mark]

Question 15

A. Alpha, beta and gamma radiations have various depths of penetration in matter and different charges.

Which row in the table best shows the penetration and charge of each radiation?

	Alpha	Beta	Gamma
A	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium no charge	not fully absorbed by several mm of lead no charge
B	absorbed by several mm of aluminium positive charge	not fully absorbed by several mm of lead negative charge	absorbed by a sheet of card no charge
C	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium positive charge	not fully absorbed by several mm of lead no charge
D	absorbed by a sheet of card positive charge	absorbed by several mm of aluminium negative charge	not fully absorbed by several mm of lead no charge

[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- Alpha, beta and gamma have different properties as shown in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	-1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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- Alpha and beta radiation would be entirely stopped by a few mm of lead
- The emission of gamma radiation would only be reduced as some would still be detected through 2 mm of lead
 - Therefore, option **D** is correct

2

The correct answer is **B** because:

- Nuclei are made up of nucleons; protons and neutrons
- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and antineutrino ($\bar{\nu}_e$)
 - $${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0\beta + \bar{\nu}_e$$
 - The mass is not changed, but the proton number increases by one, the nucleon number stays the same

3

The correct answer is **D** because:

- The gold foil experiment directed parallel beams of α -particles at a piece of gold

foil only 10^{-6} m thick

- The observations were as follows:
 - Most of the α -particles went straight through the foil
 - Some of the α -particles were deflected slightly
 - Some, about 1 in 20 000, were deflected by more than 90° , appearing to bounce back off the foil
- These results confirmed that the atom was made up of empty space and that the nucleus only occupied a very small fraction of the atom

A is incorrect as the results obtained do not confirm the presence of both protons and neutrons

B is incorrect as Bohr first proposed the presence of electrons in orbit in 1913

C is incorrect as the evidence from the gold foil experiment did not support the presence of **isotopes**

4

The correct answer is **A** because:

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
- When the ${}^{27}_{13}\text{Al}$ is bombarded with these particles the nuclei increases in mass and proton number as shown in the equation
 - ${}^{27}_{13}\text{Al} + {}^4_2\alpha \rightarrow {}^{30}_{15}\text{P} + {}^Z_AX$
- $27 + 4 = 30 + Z$
 - So, $Z = 1$
- $13 + 2 = 15 + A$
 - So, $A = 0$
- Therefore, 1_0X is a neutron

5

The correct answer is **B** because:

- The symbol ${}^{133}_{55}\text{Cs}$ give the mass number of 133 and an atomic number of 55
 - So, the nucleus contains 55 protons
- As the atom is neutral the number of electrons will be 55

- The mass of the atom is calculated by adding the number of neutrons to protons
 - $133 - 55 = 78$ neutrons

6

The correct answer is **C** because:

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- α -particles have a slower speed than that of gamma or beta due to its mass
 - The typical speed of an α -particle is 10^6 m s^{-1} to 10^7 m s^{-1}

A is incorrect as the energy of an α -particle changes as the particle moves away from the nucleus it was emitted from

B is incorrect as α -particles have a higher ionising power than β -particles

D is incorrect as the charge on an α -particle is $+2e$, a charge of 3.2×10^{-19}

7

The correct answer is **A** because:

- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}^0_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one
- Therefore, nucleon numbers are unchanged by the emission of β -particles

B is incorrect as in a nuclear reaction, the total energy is conserved. The apparent loss of mass is conserved in kinetic energy

C is incorrect as an atom's mass number can never be smaller than its atomic number

D is incorrect as if the atom has a different number of protons, then it is a different atom

8

The correct answer is **B** because:

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
 - ${}^{258}_{92}U \rightarrow {}^{254}_{90}X + {}^4_2\alpha$
- So, the proton number decreases by two
 - Therefore, atom X has 90 protons
- So, nucleon number will reduce by four
 - Therefore, atom X has a nucleon number of 254

9

The correct answer is **A** because:

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}^0_{-1}\beta + \bar{\nu}_e$
 - The mass is not changed, but the proton number increases by one
- The original nucleus Q will contain ${}_A^ZX$ protons and neutrons
 - $Q \rightarrow R + {}^4_2\alpha + 2 {}^0_{-1}\beta$
- Isotopes are nuclei with the same proton number and a different number of neutrons and therefore mass
 - So, alpha decay will decrease the proton number by 2
- Then, beta decay will turn two neutrons into two protons and two beta particles
 - There is no change to the proton number, so, R is an isotope of Q

B is incorrect as there will be a change in the nucleon number as the nucleon will have decreased by four in R.

C is incorrect as beta decay occurs when a neutron is changed to a proton

D is incorrect as there will be a mass defect which results in the release of energy so the total mass will not be conserved

10

The correct answer is **B** because:

10

The correct answer is **B** because:

- When E absorbs a neutron the new isotope is ${}_{14}^{29}\text{E}$
- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- When the new isotope emits an alpha particle, the nucleon number is reduced by 4
 - So, the new nucleon number is $29 - 4 = 25$
- The proton number is reduced by 2
 - So, the new proton number is $14 - 2 = 12$
- Therefore, the new atom (${}_{12}^{25}\text{M}$) is nuclei M

A is incorrect as 24 is the nucleon or mass of atom Q, the number of electrons in an uncharged atom of Q will be 13

C is incorrect as nuclei G and M have different proton numbers making them different atoms not isotopes

D is incorrect as in β^- decay, a neutron decays to become a neutron and proton. The mass is not changed, but the proton number increases by one. Nucleus M after β decay would be ${}_{13}^{25}\text{X}$

11

The correct answer is **A** because:

- The gold foil experiment directed parallel beams of α -particles at a piece of gold foil only 10^{-6} m thick
- The observations made in the experiment were as follows:
 - Most of the α -particles went straight through the foil
 - Some of the α -particles were deflected slightly
 - Some, about 1 in 20 000, were deflected by more than 90° , appearing to bounce back off the foil
- In order to increase the proportion of α -particles that are deflected by a larger angle, more nuclei would need to be present
 - Thicker foil would mean that more nuclei are present

B is incorrect as fewer particles in the nuclei would have less of a deflection

C is incorrect as the number of nuclei is not changing so the kinetic energy of the α -particle will have no effect

D is incorrect as the number of nuclei is not changing, so the higher activity of the α -particle will have no effect.

12

The correct answer is **D** because:

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- The nucleon number is reduced by 4 making this 235
- The proton number is reduced by 2 making this 92
 - ${}^{239}_{94}\text{Pu} \rightarrow {}^{235}_{92}\text{X} + {}^4_2\alpha$

13

The correct answer is **C** because:

- Isotopes are nuclei with the same proton number and a different number of neutrons and therefore mass
- The first nucleus has a nucleon number of 186 and 112 neutrons, so, this will make a total of $186 - 112 = 74$ protons
- The second nucleus has a nucleon number of 182 and 108 neutrons, so, this will make a total of $182 - 108 = 74$ protons
 - They have the same number of protons, therefore, they are isotopes of each other

A is incorrect as	$186 - 112 = 74$
	$180 - 118 = 62$
	these nuclei are not isotopes of each other
B is	$184 - 110 = 74$
	$187 - 110 = 77$

incorrect as these nuclei are not isotopes of each other

D is
incorrect as

$$186 - 110 = 76$$

$$186 - 112 = 74$$

these nuclei are not isotopes of each other

14

The correct answer is **C** because:

- Alpha, beta and gamma have different properties as shown in the table below
- Alpha radiation has the greatest ionising power

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	-1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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15

The correct answer is **D** because:

- Alpha, beta and gamma have different properties as shown in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	-1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - It is absorbed by card and has a positive charge
- A β^- particle is made of an electron, ${}^0_{-1}\beta$
 - It is absorbed by aluminium and is negatively charged

- A γ -ray is an electromagnetic wave
 - It is not fully absorbed by lead and has no charge

